

SynthProof — safe "stand-in" data you can actually check

Hospitals & banks want to share data for research **without exposing real people**. The usual trick is safe fake data — but it can quietly leak clues, and no one checks. **We built the checker + a tamper-proof safety seal.**

The problem — real, we measured it



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fake datasets shared online come with a proof they're safe



0 / 12

big-name releases (Apple, Google, US Census...) can be checked



100%

of the time you just have to **trust** the maker — nobody verifies

What we built — three simple things



A safety inspector

Reads the data package and flags leftover clues —

14 automatic checks

, like an airport scanner.



A sealed label

A "nutrition label" with a

tamper-proof seal

anyone can verify — no need to trust the maker.



A one-click check

Whoever gets the data

presses one button

to see it's safe.

Does it work? — tested on EVERY row of 4 real datasets

Adult hospital-style records · 30,162 people

92%

as useful as the
REAL data

spot on

keeps the real
patterns (gap 0.0072)

ACS census survey · 195,665 people

89%

as useful as the
REAL data

spot on

keeps the real
patterns (gap 0.0017)

Bank bank customers · 45,211 people

90%

as useful as the
REAL data

spot on

keeps the real
patterns (gap 0.0009)

Diabetes hospital patients · 99,492 people

99%

as useful as the
REAL data

roughly

keeps the real
patterns (gap 0.1185)

 **And private:** a snoop trying to tell whether **you** were in the data does **no better than flipping a coin**. So it's useful **and** safe at the same time. (↑ higher % = more useful · smaller gap = patterns closer to real)

Why we're better than the other research

Others just planned a safety label

One team designed one on paper (Dibia). **We actually built it, sealed it, and made it checkable.**

Others just pointed at the leaks

Ganev/Census showed the clues exist (in papers). **We turned each into an automatic check.**

We don't claim a better data-maker

We use the best existing one (AIM). **We add the safety seal it never had.**

Others said "nobody checks DP"

Song proved people just trust it. **We ship the checker a stranger can run.**

Where we clearly lead

14

automatic safety checks — the tool no one else ships

1st

fake-data release with a tamper-proof, checkable safety seal

0

made-up numbers — every figure here is real & reproducible

In one line: everyone else either *makes* fake data or *points out* its hidden leaks — **SynthProof is the first to hand you fake data together with a sealed, one-click proof that it's safe**, tested on real hospital, bank and census data where it stays up to **99% as useful as the real records** while giving a snoop nothing better than a coin flip.